

Learner's Workbook

Worksheets and exercises for the 40-hour CDF intensive

SpaceCDF -- AI-Assisted Concurrent Design Facility

University of Ottawa - SEDTI

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DRAFT

Learner's Workbook

Worksheet 1.1: Introduction to Space Mission Design

Name:	
Date:	
Team:	
Session:	1.1 -- Introduction to Space Mission Design

Part A: The 17 SE Processes -- SpaceCDF Mapping

For each NASA SE process (NPR 7123.1D), identify which SpaceCDF feature implements it. Write the tab name, button, or panel. If the process requires human judgment that the tool cannot automate, mark "Manual" in the last column.

Hint: Navigate through SpaceCDF's workflow steps (Need -> Concept -> Requirements -> Design) and explore each tab. Many processes map to more than one feature.

#	SE Process	SpaceCDF Feature(s)	Manual
1	Stakeholder Expectations Definition		
2	Technical Requirements Definition		
3	Logical Decomposition		
4	Design Solution Definition		
5	Product Implementation		
6	Product Integration		
7	Product Verification		
8	Product Validation		
9	Product Transition		
10	Technical Planning		
11	Requirements Management		
12	Interface Management		
13	Technical Risk Management		
14	Configuration Management		
15	Technical Data Management		
16	Technical Assessment		
17	Decision Analysis		

Discussion question: Which processes are well-supported by the tool? Which ones fundamentally require human judgment and cannot be fully automated? Write your answer below:

Part B: Lifecycle Phase Identification

For each activity below, identify which NASA lifecycle phase (Pre-A through F) it belongs to and which review gate marks the end of that phase:

Activity	NASA	Review Gate
"We need to monitor sea ice thickness in the Canadian Arctic"		
Writing system-level "shall" requirements with verification methods		
Selecting a specific reaction wheel vendor after evaluating 3 options		
Running thermal-vacuum qualification tests on the flight model		
Operating the satellite and producing daily data products		
Decommissioning the satellite and performing passivation		
Preliminary orbit trade study comparing SSO, ISS-orbit, and polar		
Detailed finite element analysis of the primary structure		

Hint: Refer to the lifecycle phase table in Session 1.1, Section 4.3. Ask yourself: "What level of design detail does this activity require?"

Part C: CDF Position Interaction Analysis

For each design conflict below, identify which engineering positions must interact to resolve it. Then briefly describe how you would approach the resolution.

Conflict 1: The payload requires 50 W but the solar array can only provide 30 W.

Positions involved: _____

Resolution approach: _____

Conflict 2: The reaction wheel vibration exceeds the payload's pointing stability requirement.

Positions involved: _____

Resolution approach: _____

Conflict 3: The selected orbit (700 km) requires a propulsive deorbit for debris compliance, but there is no mass budget remaining for a propulsion system.

Positions involved: _____

Resolution approach: _____

Conflict 4: The X-band downlink antenna pattern interferes with the payload's sensitive optical receiver.

Positions involved: _____

Resolution approach: _____

Part D: SpaceCDF Tool Exploration

Navigate through the SpaceCDF tool and answer each question:

1. How many KPI cards are shown on the Design Dashboard? List them by name:

2. In the Gate Review tab, what are the MCR exit criteria? Which are auto-evaluated?

3. What engineering positions are available when starting a collaborative session?

4. What workflow steps does SpaceCDF guide you through (from left to right)?

5. In the Positions tab, find your assigned position. What key questions does it list?

Part E: V-Model Sketch

In the space below, sketch the System-V model from memory. Label each level on both the left side (decomposition) and right side (integration). Draw the horizontal traceability arrows and label them.

Use this space for your sketch:

Notes & Reflections

Record key points from the group discussion, questions you want to follow up on, and connections to your mission concept:

Worksheet 1.2: Canadian Space Ecosystem & Regulatory Environment

Name:	
Date:	
Team:	
Session:	1.2 -- Canadian Space Ecosystem & Regulatory Environment

Part A: Canadian Space Industry Mapping

For your team's mission concept, identify potential Canadian industry partners at each tier. If no Canadian supplier exists for a capability, note what country you would source from and why.

Tier	Capability Needed	Canadian Candidate(s)	International Alternative	Justification
Tier 1 (Prime/Integration)				
Tier 2 (Subsystem)				
Tier 2 (Subsystem)				
Tier 3 (Component/Service)				
Tier 4 (R&D/Academic)				

Question: What CSA funding program(s) might support your mission? Why?

Part B: RSSSA Compliance Assessment

Answer the following questions for your team's mission concept:

- Does your mission involve any remote sensing of Earth (imaging, SAR, radiometry, etc.)?

- If yes, will the satellite be operated from Canada?

- What RSSSA conditions might apply? (Check all that apply)

Condition	Applies?	Notes
Priority access for Government of Canada		
Shutter control provisions		
Data disposition / handling plan		
System disposal plan		

Raw data protection

Interoperability arrangements (foreign access)		
--	--	--

4. If your mission does NOT involve remote sensing, what other regulatory requirements apply? (e.g., spectrum licensing, export control)

Part C: Spectrum Licensing Plan

Identify the frequency bands your mission will use and the licensing path for each:

Link	Band	Frequency (GHz)	Data Rate	Use (TT&C / Payload / ISL)	Licensing Path
Uplink					
Downlink (TT&C)					
Downlink (Payload)					
Other					

Hint: For a typical CubeSat, TT&C uses S-band (2.0--2.3 GHz) and payload data uses X-band (8.025--8.4 GHz). The licensing path for each is: ISED application -> ISED technical review -> ITU filing (API) -> ITU coordination -> ISED licence issuance.

Question: How early in your project should you begin the spectrum licensing process? Why?

Part D: Export Control Classification

List the 3 most critical components in your preliminary mission concept. For each, identify the likely export control classification:

Component	Country of Origin	Likely Classification	Licence Required?
		ECCN: / USML Cat:	
		ECCN: / USML Cat:	
		ECCN: / USML Cat:	

Hint: Most commercial-off-the-shelf (COTS) space components from US manufacturers are classified under EAR (ECCN 9A515 for spacecraft parts). Military-grade or specifically designed items may fall under ITAR

(USML Category XV). Canadian-origin components are subject to EIPA if they appear on the Export Control List.

Question: If your mission includes a US-origin star tracker and you plan to launch on an Indian PSLV, what export control issues arise?

Part E: Regulatory Timeline

Based on your mission concept, estimate when each regulatory activity should begin and how long it will take. Mark the activities on the timeline below:

Regulatory Activity	Start Phase	Duration	Critical Path?
RSSSA licence application			
ISED spectrum application			
ITU coordination			
Export permit(s)			
Launch provider interface agreement			

Question: Which regulatory activity is most likely to be on the critical path for your mission? Why?

Notes & Reflections

Record key points from the discussion, regulatory questions you need to research further, and implications for your mission design:

Worksheet 1.3: International Standards for Space Systems

Name:	
Date:	
Team:	
Session:	1.3 -- International Standards for Space Systems

Part A: Standard Applicability Matrix

For your team's mission concept, determine which ECSS and NASA standards apply. For each standard, decide whether it is fully applicable, partially applicable (with tailoring), or not applicable. Justify your decision.

Standard	Branch	Applicability	Tailoring Notes / Justification
ECSS-M-ST-10C (Project Management)	M		
ECSS-M-ST-40C (Configuration Management)	M		
ECSS-M-ST-80C (Risk Management)	M		
ECSS-E-ST-10C (Systems Engineering)	E		
ECSS-E-ST-20C (Electrical & Electronic)	E		
ECSS-E-ST-32C (Structures)	E		
ECSS-E-ST-40C (Software)	E		
ECSS-E-ST-50C (Communications)	E		
ECSS-Q-ST-10C (Product Assurance)	Q		
ECSS-U-AS-10C (Debris Mitigation)	U		
ISO 24113 (Debris Mitigation)	--		
NPR 7123.1D (SE Processes)	--		

Hint: A university CubeSat mission would typically invoke ECSS-E-ST-10C with heavy tailoring, ECSS-U-AS-10C fully, and might waive ECSS-Q-ST-10C (product assurance) due to cost constraints. A CSA-funded mission would typically require more complete ECSS compliance.

Question: What is the minimum set of standards your mission MUST comply with regardless of mission class?

Part B: CDS Rev 14 Compliance Checklist

If your mission uses a CubeSat form factor, complete this compliance checklist. If not, skip to Part C.

CDS Requirement	Your Design Value	Compliant?	Notes
Unit dimensions (100 x 100 x 113.5 mm per U)			
Mass per U (≤ 2.0 kg)			
Total mass (for your form factor)			
Rail material (hard-anodised Al)			
CG within 2 cm of geometric centre			
No protrusions beyond envelope during launch			
RF silence for 30 min after deployment			
Deployment switches on all deployables			
Propulsion inhibits (3 independent, if applicable)			

Hint: Use SpaceCDF's Compliance panel to check CDS compliance. The tool will flag non-compliant parameters automatically based on your mass and dimensions entries.

Part C: Debris Mitigation Compliance

Complete the following debris mitigation assessment for your mission orbit:

- Planned orbit altitude: _____ km
- Estimated natural orbital lifetime (without propulsion):

Use the rule of thumb: $\tau_{\text{years}} \approx \frac{h - 200}{30} \times \frac{m/A}{50}$ where h is altitude (km), m is mass (kg), A is cross-section area (m^2)

$h =$ _____ km, $m =$ _____ kg, $A =$ _____ m^2

$m/A =$ _____ kg/m^2

$\tau =$ _____ years (approximate)

- Compliance assessment:

Requirement	Limit	Your Mission	Compliant?
IADC post-mission lifetime	≤ 25 years	_____	_____
FCC post-mission lifetime (2024+)	≤ 5 years	_____	_____
ISO 24113 disposal reliability	$\geq 90\%$	_____	_____

4. If non-compliant with the 5-year FCC rule, what options exist?

5. Passivation plan (list all stored energy sources and how you will deplete them at end of life):

Energy Source	Passivation Method
Battery	_____
Pressure vessels (if any)	_____
Reaction wheels	_____
RF transmitter	_____
Other: _____	_____

Part D: Structural Design Margins

If your mission has preliminary structural design information, calculate the margin of safety:

Use the formula: $\text{MoS} = \frac{\sigma_{\text{allowable}}}{\text{FoS} \times \sigma_{\text{applied}}} - 1$

Worked example: For Al 7075-T6 ($\sigma_{\text{yield}} = 503$ MPa), applied stress = 200 MPa, FoS (yield) = 1.25:

$$\text{MoS} = \frac{503}{1.25 \times 200} - 1 = \frac{503}{250} - 1 = 1.012$$

This is a positive margin (compliant).

Your calculation:

Material: _____, $\sigma_{\text{allowable}} =$ _____ MPa

$\sigma_{\text{applied}} =$ _____ MPa, FoS = _____

MoS = _____ (show work below)

Part E: SpaceCDF Compliance Panel

Navigate to the Compliance panel in SpaceCDF and answer:

1. What is the debris mitigation compliance status for your mission?

2. What orbital altitude would be needed to comply with the FCC 5-year rule WITHOUT propulsion? (Use the tool's orbital lifetime calculator)

3. What standards has the tool automatically suggested for your mission type?

4. Are there any compliance flags (warnings or errors) on your current design?

Notes & Reflections

Record key points about standards applicability, compliance challenges, and design implications for your mission:

Worksheet 1.4: Mission Needs, Trade Studies & Concept of Operations

Name:	
Date:	
Team:	
Session:	1.4 -- Mission Needs, Trade Studies & ConOps

Part A: Stakeholder Analysis Matrix

Identify at least 4 stakeholders for your team's mission. For each, capture their needs, constraints, priority, and influence level.

#	Stakeholder	Role	Key Needs	Constraints Imposed	Priority (P/S/T)	Influence (H/M/L)
1						
2						
3						
4						
5						
6						

P = Primary (must satisfy), S = Secondary (should satisfy), T = Tertiary (nice to have)

H = High influence, M = Medium, L = Low

Question: Which stakeholder has the most influence over your mission? Do any stakeholders have conflicting needs? If so, how might you resolve the conflict?

Part B: Problem Statement

Write your team's mission problem statement. It must answer all four questions: What is the problem? Who is affected? What is the impact? What constraints exist?

Problem Statement:

Peer Review Checklist (exchange with another team):

Criterion	Pass?	Feedback
Describes WHAT is needed, not HOW to achieve it?		
Identifies the capability gap?		
Names specific affected stakeholders?		
Quantifies the impact of not solving it?		
States constraints (budget, schedule, regulatory)?		
Free of solution language? (No specific satellite type, orbit, component named)		

Part C: Mission Objectives

Write 2--3 mission objectives. Score each against the quality checklist.

Objective 1:

Quality Criterion	Yes/No	Notes
States WHAT, not HOW?		
Has measurable criterion with number + unit?		
Achievable with current technology (TRL ≥ 4)?		
Relevant to the problem statement?		
Traceable to a specific stakeholder need?		
Classified as primary or secondary?		

Objective 2:

Quality Criterion	Yes/No	Notes
States WHAT, not HOW?		
Has measurable criterion with number + unit?		
Achievable with current technology (TRL ≥ 4)?		
Relevant to the problem statement?		

Traceable to a specific stakeholder need?	_____	_____
Classified as primary or secondary?	_____	_____

Objective 3 (optional): _____

Part D: MoE / MoP / TPM Trace

For your primary objective, trace the measurement hierarchy:

Objective: _____

MoE (Measure of Effectiveness -- operational effectiveness from user perspective):

MoP (Measure of Performance -- technical performance of the system):

TPM (Technical Performance Measure -- design parameter tracked during development):

Hint: If your objective is "Provide 10 m GSD multispectral imagery...", then:

- MoE might be "% of target area with usable (<30% cloud) imagery within 5 days"
- MoP might be "GSD $\leq$ 10 m at nadir, SNR $\geq$ 100 in all bands"
- TPM might be "current estimate of aperture diameter (target: 15 cm), imager mass (target: 1.5 kg)"

Part E: Trade Study -- Space vs Non-Space

Complete a trade study evaluating whether a dedicated satellite is the right solution for your mission need.

Decision statement: _____

Alternatives:

#	Alternative	Description
1		
2		
3		
4	Do nothing / status quo	

Criteria and Weights (must sum to 1.0):

Use pairwise comparison or direct assignment. Show your weight derivation below.

Criterion	Weight	Direction (higher/lower is better)
Total	1.00	

Weight derivation (pairwise comparison table or notes):

Scoring Matrix (normalised 0--1):

Alternative	Criterion 1	Criterion 2	Criterion 3	Criterion 4	Criterion 5	Weighted Total
1:						
2:						
3:						
4: Do nothing						

Recall: Weighted Total = $\sum w_c \times s(a,c)$

Result: The highest-scoring alternative is: _____

Decision statement (1 paragraph -- include top 2 alternatives and why one was selected over the other):

Sensitivity check: If you increase the weight of your second-most-important criterion by 20% (and re-normalise), does the ranking change?

Part F: Concept of Operations Outline

Document your preliminary ConOps following the NASA SEH Appendix S structure:

1. Mission Architecture:

Space segment: _____

Ground segment: _____

User segment: _____

2. Orbit-Average Power Budget:

Calculate using: $P_{\text{avg}} = \sum P_{\text{mode},i} \times DC_i$

Mode	Power (W)	Duty Cycle (fraction of orbit)	P x DC (W)
Idle / Housekeeping	_____	_____	_____
Science / Imaging	_____	_____	_____
Downlink	_____	_____	_____
Eclipse	_____	_____	_____
Orbit-average power:			_____ W

Solar array requirement: $P_{\text{SA}} =$ _____ W

Show your calculation:

3. Data Budget:

Daily data generation: _____ GB/day

Downlink capacity per pass: _____ GB/pass

Passes per day: _____

Daily downlink capacity: _____ GB/day

Balance (downlink $\geq$ generation)? _____

Part G: SpaceCDF Tool Entry

In SpaceCDF, enter your mission need (Step 1) and concept (Step 2). Then answer:

1. In Step 1, did the tool auto-detect your mission type from the objective text? What type did it suggest?

2. In Step 2, run the mission trade analysis. Does the tool's recommendation match your manual trade study result from Part E?

3. In the ConOps tab, build your architecture diagram. How many systems did you define?

4. In the Trade Studies tab, run an orbit selection trade with at least 3 orbit options. What orbit did it recommend?

Notes & Reflections

Record key decisions made today, open questions, and what you want to investigate further:

Worksheet 2.1: The System-V and Requirements Engineering

Name:	
Date:	
Team / Mission:	
Role / Position:	

SpaceCDF Tab: Requirements

Key Equations Reference

Traceability Coverage:

$$\text{Coverage} = \frac{N_{\{\text{verified requirements}\}}}{N_{\{\text{total requirements}\}}} \times 100\%$$

Target: 100% -- every requirement needs a parent trace, a verification method, and a responsible subsystem.

Requirement Quality (SMART):

Specific -- Measurable -- Achievable -- Relevant -- Traceable

Verification Methods (ATID):

Analysis -- Test -- Inspection -- Demonstration

Part A: System-V Mapping (10 min)

For your mission, identify one element at each level of the V-model:

V-Level	Left Branch (Decomposition)	Right Branch (Verification)
Stakeholder Need		
Mission Requirement		
System Requirement		
Subsystem Requirement		

Part B: Requirement Quality Assessment (15 min)

Evaluate each requirement using the SMART checklist and NASA Appendix C criteria. Mark Pass (P) or Fail (F). Rewrite any failures.

#	Requirement	S	M	A	R	T	WHAT or	Rewrite (if needed)
1	"The spacecraft shall operate at 500 km altitude"							
2	"The system shall achieve GSD $\leq$ 10 m at nadir"							
3	"The EPS shall use triple-junction GaAs cells"							
4	"The system shall have a design lifetime of at least 3 years"							
5	"The structure shall be made of aluminium 7075"							

Part C: Requirement Hierarchy and Traceability (15 min)

For your mission, write one requirement at each level. Draw the traceability chain.

Stakeholder Objective (traces FROM): _____

Mission Requirement (MR-____): _____

System Requirement (SR-____), derived from MR above: _____

Subsystem Requirement (SSR-____), derived from SR above: _____

Traceability diagram (draw arrows showing parent-child relationships):

Part D: Splitting Compound Requirements (10 min)

Split this compound requirement into individually testable statements:

"The system shall provide 10 m GSD multispectral imagery with 5-day revisit, 24-hour data latency, and positive power margin in all operational modes including eclipse."

#	Split Requirement (one concern each)	Type (F/P/I/C/E)	Verification (A/T/I/D)	Phase
1				
2				
3				
4				
5				

Part E: SpaceCDF Requirements Generation (20 min)

1. In SpaceCDF, click "Generate from Objectives." How many requirements were generated?

 2. Use the Level filter -- count at each level:
 - Mission: _____ System: _____ Subsystem: _____
 3. Classify 5 requirements by type:
 4. Find one requirement that says HOW (implementation) not WHAT. Write it and your correction:
 - Original: _____
 - Corrected: _____
 5. Accept at least 5 requirements and reject at least 1. Record your decisions:
 6. For 5 key requirements, assign verification method and phase:
-

Notes & Reflections

Worksheet 2.2: Functional Decomposition and Allocation

Req ID	Requirement Text (abbreviated)	Type (Functional / Performance / Interface / Constraint / Environmental)	
Req ID	Action (Accept/Edit/Reject)	Rationale	
Req ID	Verification Method (A/T/I/D)	Phase (B/C/D/E)	Rationale for Method
Name:	_____		
Date:	_____		
Team / Mission:	_____		
Role / Position:	_____		

SpaceCDF Tab: Functions

Key Equations Reference

Function Types: Observe, Communicate, Navigate, Point, Power, Protect, Store, Process, Support

Universal Functions (every spacecraft):

1. Generate electrical power
2. Maintain thermal environment
3. Survive launch environment
4. Communicate with ground (TTC)
5. Dispose at end of life

Coverage Rule: Every leaf function must trace to at least one requirement. No gaps.

Allocation Rule: Multi-allocated functions create interfaces that must be explicitly managed.

Part A: Mission Function Tree (20 min)

Draw your mission's complete function tree. Use minimum 3 levels of decomposition. Include both mission-specific and universal functions.

Mission-specific functions:

```
F-001: _____
  +-- F-002: _____
    +-- F-002a: _____
    +-- F-002b: _____
  +-- F-003: _____
    +-- F-003a: _____
  +-- F-004: _____
  +-- F-005: _____
```

Universal functions:

```
F-010: Generate electrical power
  +-- F-011: _____
  +-- F-012: _____
F-020: Maintain thermal environment
  +-- F-021: _____
F-030: Survive launch environment
F-040: Communicate with ground (TTC)
F-050: Dispose at end of life
```

Part B: Function-to-Requirement Traceability (15 min)

For each leaf function, write one derived requirement with a measurable threshold:

Part C: Multi-Allocation Analysis (10 min)

Identify at least one function that is allocated to more than one subsystem:

Function: _____

Subsystem A (primary): _____

Subsystem B (contributor): _____

Interface created between A and B: _____

Who "owns" this function? _____

What interface requirements are needed? _____

Part D: Performance Criteria (10 min)

For 4 key functions, define quantitative performance criteria:

Part E: SpaceCDF Coverage Check (15 min)

After entering functions in SpaceCDF:

1. How many total functions in your tree? ____
2. How many leaf functions? ____
3. How many leaf functions have NO linked requirements (amber badge)? ____
4. List the uncovered functions and derive requirements for them:
5. Are there any functions the tool generated that are NOT appropriate for your mission? List and explain why you removed them:

Notes & Reflections

Worksheet 2.3: Orbit Selection and Mission Architecture

Function ID	Function Name	Allocated To (Subsystem)	Derived	Type
Function	Performance Criterion	Value	Unit	Source
Function ID	Function Name	Derived Requirement		
Name:				
Date:				
Team / Mission:				
Role / Position:				

SpaceCDF Tabs: Mission Architecture, Orbit Trade Advisor

Key Equations Reference

Orbital period: $T = 2\pi \sqrt{a^3/\mu}$ where $a = R_E + h$, $\mu = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$, $R_E = 6371 \text{ km}$

Orbital velocity: $v = \sqrt{\mu/a}$

Eclipse fraction: $f_{\text{ecl}} = \frac{1}{\pi} \arccos\left(\frac{\sqrt{a^2 - R_E^2}}{a}\right)$

Sun-synchronous inclination: $\cos(i) = -\frac{2\dot{\Omega} a^{7/2}}{3 R_E^2 J_2 \sqrt{\mu}}$ where $J_2 = 1.0826 \times 10^{-3}$

Hohmann ΔV : $\Delta V_1 = \sqrt{\mu/r_1} \left(\sqrt{2r_2/(r_1+r_2)} - 1 \right)$

Swath width: $W = 2h \tan(\theta)$

Part A: Orbital Mechanics Calculations (20 min)

Your selected orbit: Altitude $h =$ _____ km, Inclination $i =$ _____ deg, Type: _____

Show all working for each calculation:

1. Semi-major axis:

$a = R_E + h = 6371 +$ _____ $=$ _____ km $=$ _____ m

2. Orbital period:

$$T = 2\pi\sqrt{a^3/\mu} = 2\pi\sqrt{\text{_____}^3 \wedge 3.986 \times 10^{14}} = \text{_____} \text{ s} = \text{_____} \text{ min}$$

3. Orbital velocity:

$$v = \sqrt{\mu/a} = \sqrt{3.986 \times 10^{14} \wedge \text{_____}} = \text{_____} \text{ m/s}$$

4. Eclipse fraction (maximum):

$$f = \frac{1}{\pi} \arccos\left(\frac{\sqrt{a^2 - R_E^2}}{a}\right) = \frac{1}{\pi} \arccos\left(\frac{\sqrt{\text{_____}}}{\text{_____}}\right) = \text{_____}$$

5. Sunlight and eclipse time per orbit:

$$t_{\text{sun}} = T \times (1 - f) = \text{_____} \times \text{_____} = \text{_____} \text{ min}$$

$$t_{\text{ec}} = T \times f = \text{_____} \times \text{_____} = \text{_____} \text{ min}$$

6. Sun-synchronous inclination (if applicable):

$$\cos(i) = \text{_____} \rightarrow i = \text{_____} \text{ deg}$$

Does this match your selected inclination? Y / N

Part B: Orbit Trade Matrix (10 min)

From SpaceCDF's Orbit Trade Advisor, record the top 3 candidates:

Rank	Orbit Type	Alt (km)	Inc (deg)	GSD	Revisit (d)	Natural Lifetime (yr)	FCC 5-yr?	Score
1								
2								
3								

Selected orbit: _____ Rationale: _____

Part C: Debris Compliance (10 min)

For your selected orbit:

Parameter	Value
Natural orbital lifetime	_____ years
FCC 5-year compliant?	Y / N
IADC 25-year compliant?	Y / N
Propulsion needed for deorbit?	Y / N
If yes, estimated deorbit ΔV	_____ m/s
Deorbit method (if needed)	_____

Show ΔV calculation if propulsion needed:

Part D: Downstream Cascade Analysis (10 min)

How does your orbit selection affect each subsystem? Fill in the impact:

Subsystem	Orbit Parameter That Matters	Impact on Your Design
Payload (GSD)	Altitude = _____ km	GSD = _____ m
EPS (eclipse)	Eclipse fraction = _____	Eclipse duration = _____ min
Comms (link)	Slant range at 10 deg elev = _____ km	FSPL increase = _____ dB
Thermal	Eclipse/sunlight cycling	Hot case / Cold case concern: _____
AOCS	Orbit rate = _____ deg/min	Nadir tracking rate needed
Propulsion	Lifetime = _____ yr	Propulsion needed? Y / N
Radiation	TID estimate = _____ krad/yr	Electronics class: _____

Part E: Discussion (10 min)

What would happen to your mission if you moved the orbit 100 km higher?

What would happen if you moved it 100 km lower?

Notes & Reflections

Worksheet 2.4: Mission Architecture -- Segments, Interfaces, and Budgets

Name:	
Date:	
Team / Mission:	
Role / Position:	

SpaceCDF Tabs: System Architecture, Interfaces, Dashboard, Budget Breakdown

Key Equations Reference

Mass margin: $\text{Margin} \% = \frac{M_{\text{alloc}} - M_{\text{MEV}}}{M_{\text{alloc}}} \times 100\%$

Subsystem	Safe (W)	Idle (W)	Imaging (W)	Downlink (W)	Eclipse (W)
OBC					
AOCS					
Payload					
Comms (TX)					
Thermal					
Total					
Duty cycle (%)					

Orbit-average power calculation (show working):

$$P_{\text{avg}} = \text{ } \times \text{ } + \text{ } \times \text{ } + \text{ } \times \text{ } + \text{ } \times \text{ } = \text{ } \text{ W}$$

SA power required:

$$P_{\text{recharge}} = \frac{P_{\text{ecl}} \times t_{\text{ecl}}}{t_{\text{sun}} \times \eta_{\text{charge}}} = \frac{\text{ } \times \text{ }}{\text{ } \times 0.9} = \text{ } \text{ W}$$

$$P_{\text{SA,EOL}} = P_{\text{peak}} + P_{\text{recharge}} = \text{ } + \text{ } = \text{ } \text{ W}$$

$$P_{\text{SA,BOL}} = P_{\text{SA,EOL}} / (1 - 0.025)^n = \text{ } / \text{ } = \text{ } \text{ W}$$

Part C: Interface Identification (15 min)

List the 5 most critical interfaces in your design:

#	Subsystem A	Subsystem B	Interface Types (M/E/T/D/R/O)	Key Concern
1				
2				
3				
4				
5				

Write 2 formal interface requirements for your most critical pair:

Interface: _____ <-> _____

IR-001: _____

IR-002: _____

Verification method for IR-001: _____ IR-002: _____

Part D: Conflict Resolution (10 min)

From the SpaceCDF Interface Matrix, identify one conflict (red border):

Conflict description: _____

Severity: Critical / Major / Minor

Resolution options considered:

#	Option	Pros	Cons
1			
2			
3			

Selected resolution: _____

Rationale: _____

Part E: Budget Health Check (10 min)

From the SpaceCDF Dashboard, record all KPIs:

Budget	Value	Margin	Status (G/A/R)
Mass	_____ kg wet vs _____ kg alloc	_____ %	
Power (worst mode)	_____ W vs _____ W SA	_____ %	
Link	_____ dB margin	>= 3 dB?	
Cost	_____ MEUR vs _____ ceiling	_____ %	
Pointing	_____ deg vs _____ deg req	_____ %	

Which budget is tightest? _____

What single design change would improve it? _____

Notes & Reflections

Worksheet 3.1: Power System and Thermal Control Design

Name:	
Date:	
Team / Mission:	
Role / Position:	

SpaceCDF Tabs: Dashboard (Power KPI), Engineering Budgets, Timing Budget, Parametric

Key Equations Reference

SA EOL power: $P_{\text{SA,EOL}} = P_{\text{peak}} + \frac{P_{\text{ec}}}{t_{\text{ec}}} t_{\text{sun}} \eta_{\text{charge}}$

SA BOL power: $P_{\text{SA,BOL}} = \frac{P_{\text{SA,EOL}}}{(1 - \delta)^n}$ ($\delta = 0.025/\text{yr}$ for GaAs)

SA area: $A = \frac{P_{\text{BOL}}}{\eta_{\text{cell}}} \times S \times \cos\theta \times f_{\text{pack}}$
 ($\eta = 0.295$, $S = 1361 \text{ W/m}^2$, $f_{\text{pack}} = 0.85$)

[illegible]

Thermal equilibrium: $T = \left(\frac{Q_{\text{abs}}}{A_{\text{rad}}} + Q_{\text{int}} \right)^{1/4}$ ($\sigma = 5.67 \times 10^{-8} \text{ W/m}^2/\text{K}^4$)

ECSS thermal margins: Operating +/-5 degC, Acceptance +/-10 degC, Qualification +/-15 degC

Part A: Solar Array Sizing (20 min)

Show all calculations step by step.

1. Peak sunlight power demand: $P_{\text{peak}} = \underline{\hspace{2cm}}$ W (from mode:)

2. Eclipse power demand: $P_{\text{ecl}} = \quad \text{W}$

3. Eclipse and sunlight times (from Worksheet 2.3):

$$t_{\text{ec1}} = \min \quad t_{\text{sun}} = \min$$

4. Recharge power:

$$P_{\text{recharge}} = \frac{P_{\text{ecf}}}{t_{\text{ecf}}} \times t_{\text{sun}} \times \eta_{\text{charge}} = \frac{1 \times 1 \times 1 \times 1}{1 \times 0.9} = \text{_____ W}$$

5. SA EOL required:

$$P_{\{\text{SA,EOL}\}} = P_{\{\text{peak}\}} + P_{\{\text{recharge}\}} = \underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ W}$$

6. EOL degradation factor (mission lifetime = _____ years, $\Delta = 0.025/\text{yr}$):

$$(1 - 0.025)^{\{\ \ \ \ \}} = \underline{\hspace{2cm}}$$

7. SA BOL required:

$$P_{\text{SA,BOL}} = \frac{P_{\text{SA,EOL}}}{(1-\delta)^n} = \frac{\dots}{\dots} = \dots W$$

8. SA area:

$$A = \frac{P_{\text{BOL}}}{m^2} \{0.295 \times 1361 \times \cos(\quad) \times 0.85\} = \frac{\quad}{\quad} = \underline{\hspace{1cm}}$$

9. SA type needed (compare to CubeSat reference values):

- ☐ Body-mounted only (1U: ~2 W, 3U: ~7 W, 6U: ~12 W)
- ☐ Single deployable (1U: ~4 W, 3U: ~15 W, 6U: ~30 W)
- ☐ Dual deployable (3U: ~25 W, 6U: ~48 W)

10. SA mass:

$$m_{\text{SA}} = A \times \sigma = \text{_____} \times \text{_____} = \text{_____} \text{ kg}$$

(body-mounted: $\sigma = 2.5 \text{ kg/m}^2$; deployable: $\sigma = 1.5 \text{ kg/m}^2$)

Part B: Battery Sizing (10 min)

Show all calculations:

1. Eclipse energy:

$$E = P_{\text{ecl}} \times \frac{t_{\text{ecl}}}{60} = \text{_____} \times \text{_____} = \text{_____} \text{ Wh}$$

2. Battery capacity required:

$$C = \frac{E}{\text{DOD} \times \eta} = \frac{\text{_____}}{0.30 \times 0.95} = \frac{\text{_____}}{0.285} = \text{_____} \text{ Wh}$$

3. With 20% margin: $C_{\text{spec}} = \text{_____} \times 1.2 = \text{_____} \text{ Wh}$

4. Battery mass:

$$m_{\text{bat}} = \frac{C_{\text{spec}}}{E_{\text{specific}}} = \frac{\text{_____}}{150} = \text{_____} \text{ kg}$$

5. Cycle life check:

$$\text{Number of eclipses in mission} = \text{_____} \text{ orbits/day} \times 365 \times \text{_____} \text{ years} = \text{_____} \text{ cycles}$$

At DOD = 30%, Li-ion provides ~10,000 cycles. Is this sufficient? Y / N

Part C: Thermal Analysis (15 min)

1. Identify hot case and cold case for your orbit:

Parameter	Hot Case	Cold Case
Solar illumination		
Operational mode		
Internal dissipation		
Coating condition	BOL / EOL	BOL / EOL
Key concern		

2. From SpaceCDF Dashboard, record thermal predictions:

Component	Min Predicted (degC)	Max Predicted (degC)	Operating Min (degC)	Operating Max (degC)
Payload / sensor				
Battery				
OBC				
Transponder				

3. Margin check (need ≥ 5 degC):

4. If any margin fails, what thermal control action would you take?

Part D: SpaceCDF Verification (15 min)

Compare your hand calculations to SpaceCDF values:

Component	Hot margin = Max_op - Max_pred	Cold margin = Min_pred - Min_op	Pass?
Parameter	Hand Calculation	SpaceCDF Value	Difference
SA power required (BOL)	_____ W	_____ W	_____ W
Battery capacity	_____ Wh	_____ Wh	_____ Wh
Orbit-average power	_____ W	_____ W	_____ W
Eclipse duration	_____ min	_____ min	_____ min

If there are significant differences, explain why:

Notes & Reflections

Worksheet 3.2: Attitude and Orbit Control System (AOCS)

Name:	
Date:	
Team / Mission:	
Role / Position:	

SpaceCDF Tabs: Dashboard (AOCS KPI), Pointing Budget, Architecture (AOCS)

Key Equations Reference

Gravity gradient torque: $T_{\text{gg}} = \frac{3\mu}{2a^3} I_z - I_x \sin(2\theta)$
Aerodynamic torque: $T_{\text{aero}} = \frac{1}{2} \rho v^2 C_D A \cdot d_{\text{cp-cm}}$
SRP torque: $T_{\text{SRP}} = \frac{S_c}{4} A_s (1+q) \cdot d_{\text{sp-cm}}$
Magnetic torque: $T_{\text{mag}} = M \times B$
Momentum storage: $H = T_{\text{dist}} \times t_{\text{desat}}/2$

Pointing error (RSS): $\theta_{\text{total}} = \sqrt{\sum \theta_i^2}$

MTQ torque: $T_{\text{MTQ}} = m_{\text{dipole}} \times B \times \sin(\alpha)$

Part A: AOCS Architecture Selection (10 min)

Pointing requirement from your mission: _____ deg (3-sigma)

AOCS architecture selected (tick one):

- ☐ Passive magnetic (> 5 deg)
- ☐ Magnetorquers only (2--5 deg)
- ☐ Reaction wheels + MTQ (0.1--2 deg)
- ☐ Fine pointing: RW + star tracker + MTQ (< 0.1 deg)
- ☐ Very fine: RW + ST + gyro + MTQ (< 0.01 deg)

Justification for selection:

Estimated AOCS mass: _____ kg Estimated AOCS power: _____ W

Part B: Disturbance Torque Calculations (20 min)

Calculate at least 2 disturbance torques for your spacecraft and orbit. Show all working.

Spacecraft properties:

Parameter	Value	Unit
Mass		kg
Form factor		U
I _z (largest MOI)		kg m ²
I _x (smallest MOI)		kg m ²
Cross-section area (largest face)		m ²
d _{cp-cm} (CP-CM offset estimate)		m
Orbit altitude		km

1. Gravity gradient torque:

$$T_{\text{gg}} = \frac{3 \times 3.986 \times 10^{14}}{r^3} \times |I_z - I_x| =$$

Parameter	Required	Selected Product
Torque	_____ mN m	_____ mN m
Momentum	_____ mN m s	_____ mN m s
Mass	--	_____ g
Power	--	_____ W
Quantity	4 (3+1 redundancy)	

Part D: Pointing Error Budget (15 min)

Complete the RSS pointing error budget:

Error Source	Value (deg)	Value^2
Sensor accuracy		
Actuator resolution		
Alignment knowledge		
Thermal distortion		
Jitter (RW)		
Orbit knowledge		
Timing error		
Sum of squares		
RSS Total = $\sqrt{\text{sum}}$	= _____ deg	

Requirement: _____ deg

Margin: _____ deg (_____ %)

Does the pointing budget close? Y / N

Which error source dominates? _____

What action would most improve the budget? _____

Part E: SpaceCDF Verification (10 min)

Compare your hand calculations to SpaceCDF's Pointing Budget display:

Parameter	Hand Calculation	SpaceCDF	Match?
RSS pointing error	_____ deg	_____ deg	
Dominant contributor	_____	_____	
Margin to requirement	_____ %	_____ %	

If there are differences, explain:

Notes & Reflections

Worksheet 3.3: Communications and Link Budget Design

Name:	
Date:	
Team / Mission:	
Role / Position:	

SpaceCDF Tabs: Link Budget, Spectrum Selector, Equipment Browser (Comms)

Key Equations Reference

$$EIRP: \text{EIRP} = P_{TX} + G_{TX} - L_{TX} \text{ (dBW)}$$

$$FSPL: \text{FSPL} = 20 \log_{10} \left(\frac{4\pi d f_c}{c} \right) \text{ dB}$$

$$G/T: G/T = G_{RX} - 10 \log_{10}(T_{sys}) \text{ dB/K}$$

$$C/N_0: C/N_0 = \text{EIRP} - \text{FSPL} - L_{\text{losses}} + G/T + 228.6 \text{ dBHz}$$

$$E_b/N_0: E_b/N_0 = C/N_0 - 10 \log_{10}(R_b) \text{ dB}$$

$$\text{Link margin: Margin} = E_b/N_{0, \text{avail}} - E_b/N_{0, \text{req}} - L_{\text{impl}} \geq 3 \text{ dB}$$

$$\text{Antenna gain: } G = \eta_a (\pi D / \lambda)^2$$

$$\text{dB conversion: } P_{\text{dBW}} = 10 \log_{10}(P_W); \text{ } 2 \text{ W} = +3.0 \text{ dBW}; 1 \text{ W} = 0 \text{ dBW}$$

Part A: Frequency Band Selection (10 min)

Mission data rate requirement: _____ Mbps

License type selected: Amateur / Experimental / Commercial

Band selected: _____ & Centre frequency: _____ MHz

Rationale for band selection:

Licensing/filing required: _____

Estimated licensing cost: _____ & Timeline: _____ months

Part B: Complete Link Budget (25 min)

Fill in ALL rows. Show computation for EIRP, FSPL, G/T, C/N₀, and margin.

Line	Parameter	Formula / Source	Value	Unit
1	TX Power	_____ W =		dBW

2	TX Antenna Gain			dBi
3	TX Line Losses			dB
4	EIRP = Line 1 + 2 + 3		_____	dBW
5	Free Space Path Loss	$20\log_{10}(4\pi \times \text{_____} \times \text{_____} / c)$		dB
6	Atmospheric Loss			dB
7	Pointing Loss			dB
8	Polarisation Loss			dB
9	RX Antenna Gain			dBi
10	System Noise Temp	_____ K =		dBK
11	G/T = Line 9 - 10			dB/K
12	Boltzmann Constant		+228.6	dBW/K/
13	C/N ₀ = 4 + 5 + 6 + 7 + 8 + 11 + 12			dBHz
14	Data Rate	_____ bps = $10\log_{10}(\text{_____})$		dBbps
15	E _b /N ₀ available = 13 - 14			dB
16	E _b /N ₀ required (mod+code: _____)			dB
17	Implementation Loss			dB
18	LINK MARGIN = 15 - 16 - 17			dB

Show FSPL calculation:

$$\text{FSPL} = 20\log_{10}(4\pi) + 20\log_{10}(d) + 20\log_{10}(f) - 20\log_{10}(c)$$

$$= 21.98 + 20\log_{10}(\text{_____}) + 20\log_{10}(\text{_____}) - 169.54 = \text{_____ dB}$$

Does the link close (≥ 3 dB)? Y / N

If margin is excessive, maximum achievable data rate at 3 dB margin:

$$R_{b,\text{max}} = 10^{\{(E_b/N_{0,\text{avail}} - E_b/N_{0,\text{req}} - L_{\text{impl}} - 3)/10\}} \times R_b = \text{_____ Mbps}$$

Part C: Data Budget (10 min)

Daily data generation:

$$V_{\text{gen}} = R_{\text{payload}} \times t_{\text{imaging}} \times N_{\text{orbits}} \times f_{\text{compression}}$$

$$= \text{_____ Mbps} \times \text{_____ s} \times \text{_____} \times \text{_____} = \text{_____ MB/day} = \text{_____ GB/day}$$

Daily downlink capacity:

$$V_{\text{DL}} = R_{\text{DL}} \times t_{\text{contact}} \times N_{\text{passes}} \times \eta_{\text{protocol}}$$

$$= \text{_____ Mbps} \times \text{_____ s} \times \text{_____} \times \text{_____} = \text{_____ MB/day} = \text{_____ GB/day}$$

Data budget closes? ($V_{\text{DL}} \geq V_{\text{gen}}$): Y / N

If no, what is your mitigation plan?

Part D: SpaceCDF Comparison (10 min)

Open the Link Budget tab, enter your parameters, and compare:

Parameter	Hand Calculation	SpaceCDF	Difference
EIRP (dBW)			
FSPL (dB)			
C/N ₀ (dBHz)			
Link Margin (dB)			

If there are differences, explain:

Part E: Modulation and Coding Selection (5 min)

Selected modulation: _____

Selected FEC code: _____

Required E_b/N₀ at BER 10⁻⁶: _____ dB

Spectral efficiency: _____ bps/Hz

Rationale:

Notes & Reflections

Worksheet 3.4: Structure, Propulsion, and Equipment Selection

Name:	
Date:	
Team / Mission:	
Role / Position:	

SpaceCDF Tabs: Equipment Browser, Dashboard, Trade Studies, Budget Breakdown

Key Equations Reference

Structural Margin of Safety: $\text{MoS} = \frac{\sigma_{\text{allow}}}{\sigma_{\text{design}}} \times \text{FoS} - 1 \geq 0$

 FoS: Yield 1.25, Ultimate 1.5 (metallic)

Tsiolkovsky rocket equation: $\Delta V = I_{sp} \cdot g_0 \cdot \ln(m_0/m_f)$

Propellant mass: $m_{\text{prop}} = m_{\text{dry}} \times \left(e^{\Delta V / (I_{sp} \cdot g_0)} - 1 \right)$

 $g_0 = 9.80665 \text{ m/s}^2$

Data storage: $S = V_{\text{daily}} \times N_{\text{days}} \times f_{\text{safety}}$

Fundamental frequency: $f_1 > 40$ Hz (deployer requirement)

Part A: CDS Compliance Check (10 min)

CDS Requirement	Your Design Value	Specification	Compliant?
Form factor: _____ U	Dimensions: _____ mm	_____ mm	Y / N
Maximum mass	_____ kg (wet)	_____ kg	Y / N
Rail material		Hard anodised Al	Y / N
Deployment switches	_____ (quantity)	Min 2	Y / N
RBF pin provided	Y / N	Required	Y / N
No protrusions beyond rails (stowed)	Y / N	Required	Y / N
CG within 2 cm of geometric centre	Estimated: _____ cm offset	≤ 2 cm	Y / N
Fundamental frequency	Estimated: _____ Hz	> 40 Hz	Y / N

Non-compliance items and corrective actions:

Part B: Structural Margin Calculation (10 min)

Axial launch load: _____ g Lateral launch load: _____ g

Load per rail (4 rails, equal sharing):

$$F_{\text{axial}} = \frac{m \times a_{\text{axial}} \times g_0}{4} = \frac{ \times \phantom{a_{\text{axial}}} \times 9.81}{4} = \text{ N}$$

Stress in rail (cross-section 8.5×8.5 mm = $7.225 \times 10^{-5} \text{ m}^2$):

$$\sigma = \frac{F}{A} = \frac{ \times 10^{-5}}{7.225 \times 10^{-5}} = \text{ MPa}$$

Margin of safety (yield, Al 7075-T6, $\sigma_y = 503$ MPa):

$$\text{MoS} = \frac{503}{ \times 1.25} - 1 = $$

Pass? Y / N

Part C: Propulsion Decision (15 min)

Does your mission need propulsion?

Question	Answer
Orbit altitude	_____ km
Natural orbital lifetime	_____ years
FCC 5-year compliant without propulsion?	Y / N
Orbit maintenance required?	Y / N
Constellation phasing required?	Y / N
Active deorbit required?	Y / N

Decision: Propulsion / No propulsion

If propulsion is selected, complete this section. Show all working.

Total ΔV required:

Manoeuvre	ΔV (m/s)
Orbit maintenance (_____ yr)	
Deorbit	
Collision avoidance margin	
Total	

Propulsion type selected: _____ $I_{sp} =$ _____ s

Propellant mass (Tsiolkovsky):

$$m_{\text{prop}} = m_{\text{dry}} \times \left(e^{\Delta V / (I_{sp} \times g_0)} - 1 \right)$$

$$= \text{_____} \times \left(e^{\text{_____} / (\text{_____} \times 9.81)} - 1 \right) = \text{_____} \times \left(e^{\text{_____}} - 1 \right) = \text{_____} \times \text{_____} = \text{_____} \text{ kg}$$

System dry mass (thruster + tank + feed): _____ kg

Total propulsion system mass: _____ kg (propellant + dry)

Fraction of total spacecraft mass: _____ %

Part D: Equipment Selection Log (25 min)

Using SpaceCDF's Equipment Browser, select components for each category:

Category	Component Selected	Manufacturer	Mass (kg)	Power	Cost (kEUR)	Qty	Total Mass
EPS Board						1	
Battery						1	
Solar Panels							
OBC						1	
Reaction Wheel						x	
Magnetorquer						x	
Star Tracker							
Sun Sensor						x	
Transponder						1	
Antenna							
Payload						1	
Structure						1	
Harness						1	
Propulsion							
TOTAL							

Parametric estimate (from Session 2.4): _____ kg

Equipment-based total: _____ kg

Difference: _____ kg (_____ %)

Any incompatibilities detected? (RF band mismatch, voltage mismatch, interface conflict)

Part E: Component Trade Study (15 min)

Subsystem traded: _____ Options compared: 3

Criterion	Weight	Option A: _____	Score	Option B: _____	Score	Option C: _____	Score
Mass	0.20						
Power	0.15						
Cost	0.20						
TRL	0.15						
Heritage	0.15						
Performanc	0.15						
Weighted	1.00						

Winner: _____ Score: _____

Rationale (beyond the numbers):

Part F: Final Budget Health Check (10 min)

After equipment selection, record final budget status:

Budget	Value	Margin	Status
Mass (wet vs allocation)	___ / ___ kg	___ %	G / A / R
Power (worst mode vs SA)	___ / ___ W	___ %	G / A / R
Link margin	___ dB	>= 3 dB?	G / A / R
Cost vs ceiling	___ / ___ MEUR	___ %	G / A / R
Pointing (RSS vs req)	___ / ___ deg	___ %	G / A / R
Data (DL vs gen)	___ / ___ GB/day	___ %	G / A / R

Any budget that does not close? _____

Proposed fix: _____

Notes & Reflections

Worksheet 4.1: Equipment Selection & Bill of Materials

Name: _____ Date: _____ Team: _____

Mission Name: _____ Form Factor: _____ U

Part A: Make/Buy/Reuse Decisions

For each subsystem, document the procurement decision and rationale:

Subsystem	Decision (Buy/Reuse/Make)	Rationale	Lead Time
Structure			
EPS (solar array)			
EPS (battery)			
EPS (distribution)			
OBC			
AOCS (sensors)			
AOCS (actuators)			
TTC (transceiver)			
TTC (antenna)			
Thermal			
Payload			
Propulsion (if applicable)			

Part B: Bill of Materials (BOM)

Complete the BOM from the SpaceCDF Equipment Browser selections:

Part C: Budget Tracking

After all selections, record the live budget values from the SpaceCDF Dashboard:

Item ID	Component	Manufacturer	Part Number	Qty	Mass (g)	Power (W)	Cost (kEUR)	TRL	ECCN
STR-001									
EPS-001									
EPS-002									
EPS-003									
OBC-001									
AOCS-00									
AOCS-00									
AOCS-00									
TTC-001									
TTC-002									
THM-001									
PLD-001									
HRN-001									
Budget	BOM Total	Parametric Estimate	Allocation	Margin	Margin				
Dry mass	_____ g	_____ g	_____ g	_____ g	_____				
Peak	_____ W	_____ W	_____ W	_____ W	_____				
Total					_____ %				

cost

_____ kEUR	_____ kEUR	_____
------------	------------	-------

kEUR



kEUR



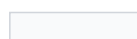
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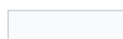
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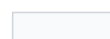
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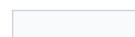


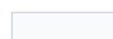
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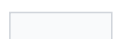


_____%









Does everything fit within allocation? Y / N

If not, what would you change? _____

Part D: Export Control Assessment

List any components that may require export licences:

Component	ECCN / USML Category	Controlled?	Licence Required?	Action Needed
		Y / N	Y / N	
		Y / N	Y / N	
		Y / N	Y / N	
		Y / N	Y / N	

Are there any ITAR-controlled items? Y / N

If yes, what TAA (Technical Assistance Agreement) or DSP-5 licence is needed?

Part E: Interface Compatibility Check

For each selected component, verify interface compatibility:

Component	Bus Voltage OK?	Data Protocol OK?	RF Band Match?	Connector OK?	Notes
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	

Part F: Interface Incompatibility Resolution

Describe one interface incompatibility found during selection and how you resolved it:

Issue: _____

Components involved: _____

Root cause: _____

Resolution chosen: _____

Impact on budget: _____

Notes & Reflections

What was the most difficult equipment selection decision? Why?

Worksheet 4.2: Verification & Validation Planning

Name: _____ Date: _____ Team: _____

Mission Name: _____

Part A: IADT Method Assignment

For 10 key requirements from your SpaceCDF V&V Matrix, assign the verification method and justify:

#	Req ID	Requirement Text (abbreviated)	Method (I/A/D/T)	Phase (B/C/D)	Level (Unit/Sub/Sys)	Justification
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

Summary count: Analysis: ____ / Test: ____ / Demonstration: ____ / Inspection: ____

Part B: Dual-Method Requirements

Identify 3 requirements that need BOTH analysis (early confidence) AND test (final proof):

Part C: Environmental Test Sequence

For your selected launch vehicle, specify the complete proto-flight test sequence:

Launch vehicle: _____ PUG reference: _____

Requirement ID	Requirement Text	Analysis (Phase B)	Test (Phase C/D)	Why Both Needed?
Step	Test	Specification	Duration	Pass Criteria
1	Initial functional test	Full performance baseline	_____ hr	All parameters nominal
2	Sine vibration	_____ g, _____ - _____ Hz	_____ min/axis x 3	No resonance shift > 5%
3	Random vibration	_____ gRMS overall	_____ min/axis x 3	No structural damage
4	Post-vibe functional test	Same as step 1	_____ hr	Compare to baseline
5	Thermal vacuum -- hot	+ _____ C (predicted max + 10 C)	_____ cycles	Functional at hot extreme
6	Thermal vacuum -- cold	- _____ C (predicted min - 10 C)	_____ cycles	Functional at cold extreme
7	Post-TVAC functional test	Same as step 1	_____ hr	Compare to baseline
8	EMC test	Required? Y / N	_____ day	No interference
9	Mass measurement	Calibrated scale (+/- 0.01 kg)	30 min	M <= _____ kg
10	Deployer fit check	_____ deployer	2 hr	Fits; slides freely

Total test campaign duration estimate: _____ weeks

Part D: Vibration Test Profile

Record the random vibration PSD profile from your launch vehicle PUG:

Frequency (Hz)	PSD Level (g^2/Hz)	Slope
20		Start
20 - _____		+ _____ dB/oct
_____ - _____		Flat
_____ - 2000		- _____ dB/oct
2000		End

Overall gRMS (calculated): _____ gRMS

Show your calculation:

Part E: TVAC Cycle Profile

Sketch or describe one TVAC cycle:

Hot extreme: + _____ C Cold extreme: - _____ C

Ramp rate: _____ C/min Dwell time at extreme: _____ hr

Functional test at each extreme? Y / N Duration of functional test: _____ hr

Calculate time for one cycle:

$t_{\text{cycle}} =$ _____

$=$ _____ hours

Total TVAC campaign: _____ cycles x _____ hr + _____ hr setup = _____ hr = _____ days

Part F: Waiver Identification

Are there any requirements that cannot be verified by the standard method? Document potential waivers:

Notes & Reflections

Which environmental test do you consider most critical for your mission? Why?

Worksheet 4.3: Risk, Interfaces & FMECA

Name: _____ Date: _____ Team: _____

Mission Name: _____

Part A: Risk Register

Identify at least 5 technical risks for your mission design. Score each on the 5x5 matrix.

Scoring reminder: $L \times C = \text{Score}$. Low (1-4), Medium (5-9), High (10-15), Critical (16-25).

Req ID	Standard Method	Proposed Alternative	Justification	Risk				
#	Risk Description	L (1-5)	C (1-5)	Score	Category (L/M/H/C)	Mitigation Strategy	Owner (Position)	Post-Mitigation Score
1								
2								
3								
4								
5								
6								
7								

How many risks are High or Critical (before mitigation)? _____

How many remain High or Critical after mitigation? _____

Part B: 5x5 Risk Matrix Plot

Plot your risks on the matrix below. Write the risk number in the appropriate cell.

		Consequence ->				
		1	2	3	4	5
L	5					
i	4					
k	3					
e	2					
l	1					

Part C: N-Squared (N^2) Interface Matrix

Fill in the interface matrix for your mission's subsystems. Use abbreviations:

- P = Power (electrical)
- D = Data (I2C, SPI, UART, CAN, RS-422)
- R = RF (coaxial)
- M = Mechanical (mounting, alignment)
- T = Thermal (conductive/radiative)

	EPS	OBC	AOCS	TTC	Payload	Structure
EPS	---					
OBC		---				
AOCS			---			
TTC				---		
Payload					---	
Structure						---

Total number of interfaces identified: _____

Interface conflicts detected: (describe at least 2)

1. Conflict: _____

Resolution: _____

2. Conflict: _____

Resolution: _____

Part D: FMECA Table

Complete for 4 critical components:

Part E: Single-Point Failure Analysis

List all single-point failures in your design:

Component	Function	Failure Mode	Cause	Local Effect	System Effect	Severity (1-5)	Detection Method	Compensating Provision
#	Component	Failure Mode	Effect on Mission	SPF?	Acceptable?	Mitigation (if not acceptable)		
1				Y/N	Y/N			
2				Y/N	Y/N			
3				Y/N	Y/N			
4				Y/N	Y/N			
5				Y/N	Y/N			

Total SPFs identified: _____ Accepted: _____ Mitigated: _____

Part F: Reliability Calculation

Calculate the series reliability for your mission's critical chain (components that must all work for mission success):

$R_{\text{series}} = R_1 \times R_2 \times \dots \times R_n =$ _____

$R_{\text{series}} =$ _____

If you added redundancy to one component (e.g., dual deployment mechanism), recalculate:

$R_{\text{redundant}} = 1 - (1 - R)^2 = 1 - (1 - \text{_____})^2 =$ _____

New R_{series} (with redundancy) = _____

Improvement: from _____ to _____ (+ _____ percentage points)

Part G: Risk Discussion

What is the highest-risk item in your design? What would it take (cost, mass, schedule) to bring the risk to an acceptable level?

Notes & Reflections

Which single-point failure concerns you the most? Would you accept it, or is mitigation essential?

Worksheet 4.4: Cost Estimation & Design Review

Name: _____ Date: _____ Team: _____

Mission Name: _____

Part A: Mission Cost Estimate (WBS)

Complete using both parametric estimates (from SpaceCDF Cost tab) and bottom-up estimates (from BOM vendor quotes):

Compon	Individual Reliability (R _i)	Mission Duration Used			
WBS #	Element	Parametric (kEUR)	Bottom-Up (kEUR)	Difference	Notes
1.0	Programme Management				
2.0	Systems Engineering				
3.0	Mission Assurance				
4.0	Payload				
5.0	Spacecraft Bus Hardware				
	5.1 Structure & mechanisms				
	5.2 EPS				
	5.3 AOCS				
	5.4 TTC				
	5.5 OBC & data handling				
	5.6 Thermal				
	5.7 Propulsion				
	5.8 Harness				
6.0	Integration & Test				
7.0	Software (FSW + GSW)				
8.0	Launch Services				
9.0	Ground Segment				
10.0	Operations (____ years)				
	TOTAL	_____	_____		

Part B: Confidence Levels

P50 estimate (your best estimate): _____ kEUR

P70 estimate (P50 x 1.2): _____ kEUR

P80 estimate (P50 x 1.3): _____ kEUR

Show your reasoning for the multiplier (is 1.3 appropriate, or should it be higher for your mission?):

Part C: Constellation Cost (if applicable)

Number of satellites: _____ Learning rate: _____%

$b = \ln(\text{learning_rate}) / \ln(2) = \ln(\text{_____}) / \ln(2) = \text{_____}$

Unit #	Unit Cost (kEUR) = $C_1 \times N^b$	Cumulative (kEUR)
1		
2		
4		
8		
_____ (total)		

Show your calculation for the total constellation cost:

Hardware: _____ units x _____ kEUR avg = _____ kEUR

Spares (_____%): _____ units x _____ kEUR = _____ kEUR

Launch: _____ sats x _____ kEUR = _____ kEUR

Ground segment: _____ kEUR

Operations (_____ years): _____ kEUR

PM/SE/MA: _____ kEUR

Total constellation cost: _____ kEUR = _____ MEUR

P80 constellation cost (x 1.3): _____ MEUR

Part D: Cost Drivers

Top 3 cost drivers in your mission (from SpaceCDF Cost Breakdown):

Rank	Cost Element	Cost (kEUR)	% of Total	Why is it Expensive?
1				
2				
3				

What would you change to reduce total cost by 20%?

Part E: Design Review Readiness

From the SpaceCDF Gate Review tab, assess your readiness for SRR/PDR:

#	Gate Criterion	Status (Pass/Fail/Manual)	Evidence Location	If Fail: Action Needed
1	Mission need justified			
2	Requirements complete & traceable			
3	All budgets close with margin			
4	Equipment selected (BOM complete)			
5	Interfaces defined, conflicts resolved			
6	V&V approach defined			
7	Risks identified with mitigations			
8	Cost estimate with WBS			
9	ConOps defined			
10	Sustainability compliance			

Must-pass criteria met: ____ / ____

Overall readiness: READY / NOT READY

Action items to achieve readiness:

1. _____
2. _____
3. _____

Part F: Presentation Preparation

Outline your 12-minute design review presentation:

Section	Duration	Key Message	Evidence/Data to Show
1. Mission Need	2 min		
2. Why Space?	2 min		
3. System Design	3 min		
4. Equipment & V&V	2 min		
5. Risk & Cost	2 min		
6. Operations & Lessons	1 min		

Notes & Reflections

What is the most uncertain element in your cost estimate? How would you reduce the uncertainty?

Worksheet 5.1: Ground Segment & Operations Architecture

Name: _____ Date: _____ Team: _____

Mission Name: _____ Orbit: _____ km, _____ deg inclination

Part A: Ground Station Contact Analysis

Calculate contact time and data volume per pass for your mission.

Input Parameters

Parameter	Value	Source
Orbital altitude (h)	_____ km	SpaceCDF orbit selection
Minimum elevation angle (epsilon)	_____ deg	Ground station spec (typically 5-10)
Downlink data rate (R)	_____ Mbps	SpaceCDF link budget
Protocol efficiency (eta)	_____ (typically 0.85-0.95)	CCSDS overhead estimate
Ground station latitude	_____ deg	Selected station
Earth radius (R_E)	6371 km	Constant

Calculations

Maximum slant range:

$$\rho = R_E \times (\sqrt{(h/R_E + 1)^2 - \cos^2(\epsilon)} - \sin(\epsilon))$$

$$\rho = 6371 \times (\sqrt{((\text{____})/6371 + 1)^2 - \cos^2(\text{____})} - \sin(\text{____}))$$

$$\rho = \text{____ km}$$

Average pass duration (approximate):

$$T_{\text{avg}} \sim \text{____ minutes (from STK/SpaceCDF or estimate: } \sim 6 \text{ min for 500 km SSO)}$$

Data volume per pass:

$$V_{\text{data}} = R \times T \times \eta = \text{____ Mbps} \times \text{____ s} \times \text{____} = \text{____ Mbit} = \text{____ MB}$$

Part B: Data Budget Closure

Parameter	Value	Unit
Daily data generation (payload)		MB/day
Daily data generation (housekeeping)		MB/day
Total daily generation		MB/day
Passes per day (GS 1: _____)		passes
Data per pass (GS 1)		MB
Subtotal downlink (GS 1)		MB/day
Passes per day (GS 2: _____)		passes
Data per pass (GS 2)		MB
Subtotal downlink (GS 2)		MB/day
Total daily downlink capacity		MB/day
Data budget margin		MB/day
Margin percentage		%

Does the data budget close? Y / N

If not, what changes would close it? (check all that apply)

- ☐ Add another ground station at: _____
- ☐ Increase data rate to: _____ Mbps (requires: _____)
- ☐ Reduce payload data generation to: _____ MB/day (impact: _____)
- ☐ Add onboard compression (ratio: _____:1)
- ☐ Use SatNOGS network for additional passes
- ☐ Other: _____

Part C: Ground Station Network Design

Station #	Location	Latitude	Antenna	Band	Data Rate	Passes/Day	Role
1							Primary TTC

2							Payload DL
3							Backup/SatNOGS

Total ground segment cost estimate: _____ kEUR

Cost Element	Annual Cost (kEUR)	Notes
Antenna rental / ownership		
MCS software licence		
Network connectivity		
Operations staff (_____ FTE)		
Total annual		
Mission lifetime (_____ yr)		

Part D: LEOP Timeline

Construct the LEOP timeline for the first 72 hours after separation:

Time (UTC)	Sim Time	Activity	Success Criterion	Status
T+0	Separation	Deployment switches release		[]
T+30 min	Timer expires	Antenna deployment commanded		[]
T+_____		Beacon acquisition	Carrier lock	[]
T+_____		First HK telemetry	Data decoded	[]
T+_____		First uplink command	ACK received	[]
T+_____		SA deployment (if separate)	Power generation confirmed	[]
T+_____		ADCS initialisation	Attitude determination active	[]
T+_____		ADCS calibration	Pointing < _____ deg	[]
T+_____		Full duplex validation	Uplink + downlink at ops rate	[]
T+_____		Health assessment complete	All subsystems nominal	[]

Part E: Mission Operations Timeline (Gantt Chart)

Sketch the operations timeline from launch to end of life:

Phase:	LEOP	Commission	Early Ops	Nominal Operations	EOL
Week:	1 2 3 4 5 6 7 8 9 ... 26 ... 52 ... 104 ... 156				
Staffing:	24/7	16/7	8/5	8/5 or automated	8/5
Key milestones:					
L+_____:	First light				
L+_____:	Commissioning complete				
L+_____:	Orbit maintenance manoeuvre (if applicable)				
L+_____:	End of nominal mission				
L+_____:	Extended mission (if applicable)				

L+____: Passivation and deorbit

Notes & Reflections

What is the most challenging aspect of your ground segment design? What would you do differently with more budget?

Worksheet 5.2: Mission Operations Concepts

Name: _____ Date: _____ Team: _____

Mission Name: _____

Part A: Operational Modes

Define all operational modes for your spacecraft. Use the SpaceCDF ConOps Editor to enter these.

Verify: Does the power budget close in every mode?

Mode	Description	Power (W)	Data Rate	Entry Condition	Exit Condition
Safe					
Detumble					

Standby					
Science					
Downlink					
Eclipse					
Manoeuvre					
Mode	Power Consumption (W)	Power Available (W)	Margin	Closes?	
Safe				Y / N	
Detumble				Y / N	
Standby				Y / N	
Science				Y / N	
Downlink				Y / N	
Eclipse (battery only)				Y / N	

Part B: FDIR Rules

Define at least 5 FDIR rules for your spacecraft:

#	Fault Description	Detection Method	Thresho	FDIR Level (0-4)	Autonomous	Recovery
1						
2						
3						
4						
5						
6						

Can Safe Mode sustain the spacecraft indefinitely (power + thermal)? Y / N

If not, what is the maximum Safe Mode duration? _____ hours

Can every autonomous FDIR action be overridden from ground? Y / N

If not, which cannot? _____

Part C: Nominal Procedure

Write one nominal procedure for your mission (science observation or data downlink):

Procedure ID: _____ Title: _____

Preconditions:

Step	Action	Expected Result	If Not Achieved
1			
2			
3			
4			
5			
6			
7			
8			

Post-conditions:

Part D: Contingency Procedure

Write one contingency procedure (e.g., safe mode recovery, communication loss recovery):

Procedure ID: _____ Title: _____

Trigger condition:

Severity level: Green / Yellow / Orange / Red

Step	Action	Expected Result	If Not Achieved	Timeout
1				
2				
3				
4				
5				
6				

Escalation: If this procedure fails, escalate to: _____

Post-conditions:

Part E: Operations Staffing

Estimate the staffing requirements for each mission phase:

Phase	Duration	Shifts/Day	Staff/Shift	Total FTE	Annual Cost (kEUR)
-------	----------	------------	-------------	-----------	--------------------

LEOP	_____ days	3 (24/7)			
Commissioning	_____ weeks	2 (16/7)			
Early Operations	_____ months	1 (8/5)			
Nominal Operations	_____ years	1 (8/5) or auto			
End of Life	_____ weeks	1 (8/5)			
Total					

Part F: Anomaly Scenario

Hypothetical scenario: Your satellite enters Safe Mode during a weekend. The next ground contact is Monday morning (40 hours away). Walk through your response:

1. What is the spacecraft doing in Safe Mode?

2. Will it survive 40 hours? (Check power and thermal)

3. What telemetry will you review first on Monday?

4. What is your recovery plan?

Notes & Reflections

How much autonomy should your spacecraft have? Where is the line between onboard decision-making and ground control?

Worksheet 5.3: Mission Simulation Day

Name: _____ Date: _____ Team: _____

Mission Name: _____

Your Role: _____ (FD / SC / GSO / PO / FD-nav / Logger)

Simulation Event Log

Record ALL events, decisions, and actions in real-time during the simulation:

#	Sim Time	Real Time	Event / Observation	Action Taken	Result	Logged By
1	T+0		Separation			
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						

LEOP Checklist

Check off each LEOP milestone as it is achieved during the simulation:

#	Milestone	Time Achieved	Status	Notes
1	Separation confirmed		☐	
2	30-min timer expired		☐	
3	Antenna deployed		☐	
4	Beacon acquired		☐	
5	First HK telemetry decoded		☐	
6	First uplink command ACK'd		☐	
7	Solar array deployed (if sep.)		☐	
8	ADCS detumble complete		☐	
9	ADCS calibration complete		☐	
10	Full duplex validated		☐	
11	All subsystems nominal		☐	

Anomaly Response Form 1

Complete this form for each injected anomaly during the simulation.

Field	Entry
Anomaly ID	ANO-SIM-001
Sim Time	
Real Time	
Severity	Green / Yellow / Orange / Red
Affected Subsystem	

Observation (exact telemetry values and symptoms):

Initial Assessment (what do you think happened?):

Diagnosis (root cause after investigation):

Action Taken (commands sent, procedures executed):

Step	Time	Action	Expected Result	Actual Result
1				
2				
3				
4				
5				

Final Result (system state after response):

Follow-up Required:

Procedure used: _____ Did it work as written? Y / N / Partially

If not, what changes are needed? _____

Anomaly Response Form 2

Field	Entry
Anomaly ID	ANO-SIM-002
Sim Time	
Real Time	
Severity	Green / Yellow / Orange / Red
Affected Subsystem	

Observation (exact telemetry values and symptoms):

Initial Assessment (what do you think happened?):

Diagnosis (root cause after investigation):

Action Taken (commands sent, procedures executed):

Step	Time	Action	Expected Result	Actual Result
1				
2				
3				
4				
5				

Final Result (system state after response):

Follow-up Required:

Procedure used: _____ Did it work as written? Y / N / Partially

If not, what changes are needed? _____

Anomaly Response Form 3

Field	Entry
Anomaly ID	ANO-SIM-003
Sim Time	
Real Time	
Severity	Green / Yellow / Orange / Red
Affected Subsystem	

Observation (exact telemetry values and symptoms):

Initial Assessment:

Diagnosis:

Action Taken:

Step	Time	Action	Expected Result	Actual Result
1				
2				
3				
4				

Final Result:

Procedure used: _____ Did it work as written? Y / N / Partially

Commissioning Status

After the commissioning phase of the simulation, record subsystem checkout status:

Subsystem	Checkout Complete?	Performance vs Spec	Issues Found
EPS	[]	Nominal / Degraded / Failed	
OBC	[]	Nominal / Degraded / Failed	
AOCS	[]	Nominal / Degraded / Failed	
TTC	[]	Nominal / Degraded / Failed	
Thermal	[]	Nominal / Degraded / Failed	
Payload	[]	Nominal / Degraded / Failed	
Propulsion	[]	Nominal / Degraded / Failed / NA	

Nominal Operations Summary

After the nominal operations phase, record:

Metric	Planned	Actual (Simulated)	Variance
Science observations completed			
Data downlinked (MB)			

Anomalies encountered			
Ground contacts made			
Safe mode entries			

Simulation Debrief

What went well?

What could be improved?

Design weaknesses revealed by simulation:

Procedure changes needed:

FDIR gaps identified:

Notes & Reflections

What was the most stressful moment during the simulation? What did it teach you about operations?

Worksheet 5.4: Final Review & Presentations

Name: _____ Date: _____ Team: _____

Mission Name: _____

Part A: Final Design Summary

Record the final state of your design from the SpaceCDF Dashboard:

Parameter	Value	Unit	Margin	Margin %	Status (G/A/R)
Dry mass		kg			
Wet mass (if propulsion)		kg			
SA power (EOL)		W			
Battery capacity		Wh			
Link margin (worst case)		dB			
Pointing accuracy		deg			
Data balance		MB/day			
Total cost (P50)		kEUR			
Total cost (P80)		kEUR			
Debris compliance score		/100			
Mission lifetime		years			

Do all budgets close? Y / N If not, which fail? _____

Number of unresolved conflicts: _____

Part B: Documentation Checklist

Tick each document generated from SpaceCDF Exports tab:

Document	Generated?	Complete?	Notes
MRD (Mission Requirements Document)	☐	☐	
Technical Specification	☐	☐	
Verification Plan	☐	☐	
ConOps Document	☐	☐	
SEMP (Systems Engineering Mgmt Plan)	☐	☐	
IRD (Interface Requirements Document)	☐	☐	
Risk Management Plan	☐	☐	
Test Plan	☐	☐	
Bill of Materials	☐	☐	
ITU Filing Template	☐	☐	
RSSSA Filing	☐	☐	
Export Control Assessment	☐	☐	
End-of-Life Report	☐	☐	
COPUOS Registration	☐	☐	

Total documents generated: _____ / 14

Part C: Presentation Outline

Prepare your 12-minute final design review presentation:

Section	Time	Key Points to Cover	Evidence / Data to Show
1. Mission Need	2 min		
2. Why Space?	2 min		
3. System Design	3 min		
4. Equipment & V&V	2 min		
5. Risk & Cost	2 min		
6. Operations & Lessons	1 min		

Presenter(s): _____

Backup slides prepared on: _____

Part D: Peer Review Evaluation

Complete this section while reviewing another team's presentation.

Team being reviewed: _____

Reviewers: _____

#	Criterion	Score (0-2)	Comments / Questions
1	Mission need clearly justified		
2	Requirements complete and traceable		
3	All budgets close with margin		
4	Equipment selected with justified trades		
5	Interfaces defined, conflicts resolved		
6	V&V approach defined		
7	Risks identified with mitigations		
8	Cost estimate with WBS and P80		
9	ConOps and ground segment designed		
10	Sustainability compliance		
	TOTAL	/ 20	

Presentation quality:

Criterion	Score (1-5)	Comments
Clarity of communication		
Use of evidence (data, not assertions)		
Ability to answer questions		
Time management		

Board recommendation: GO / NO GO / GO WITH ACTIONS

Action items for the presenting team:

#	Action	Responsible	Deadline
1			
2			
3			

Strongest aspect of this team's design:

Biggest area for improvement:

Part E: Lessons Learned

1. What was the most challenging design decision your team faced?

2. What would you do differently if starting the course over?

3. What does the design need next (Phase B activities)?

4. What was the most valuable SpaceCDF feature?

5. What feature was missing that you needed?

6. What concept from this course will you apply in your future work?

Part F: Self-Assessment Rubric

Rate your own learning across the three weeks of the course. Be honest -- this is for your own reflection, not grading.

Competency	Week 1 Start (1-5)	Now
------------	--------------------	-----

Define a space mission starting from the problem, not the solution			
Write verifiable, traceable requirements (SMART format)			
Conduct a structured trade study with weighted criteria			
Size subsystems using parametric models and engineering budgets			
Select COTS equipment and verify interface compatibility			
Construct a verification matrix with IADT methods			
Assess risk using a 5x5 matrix and define mitigations			
Estimate mission cost using parametric and bottom-up methods			
Design a ground segment and calculate data budget			
Define operational modes, FDIR rules, and procedures			
Respond to anomalies using a systematic process			
Present a design to a review board with evidence			

Scale: 1 = No familiarity, 2 = Basic awareness, 3 = Can apply with guidance, 4 = Can apply independently, 5 = Can teach others

Overall self-assessment:

At the start of this course, I would describe my systems engineering ability as:

Now, I would describe it as:

One thing I am most proud of from this course:

Notes & Reflections

Final thoughts on the course, the design process, and what comes next:
